Experiment 2

Website, Email, WHOIS, DNS, and Network Footprinting

1. Aim

To conduct deep technical footprinting on a target network (example.com) to extract infrastructural details including domain registration, DNS records, email routing configurations, and network topology.

> **Disclaimer:** This is a simulated lab report. All domain names and IP addresses (e.g., 198.51.100.x) are reserved for documentation and educational purposes.

2. Theoretical Background

Technical footprinting shifts from open-source web searching to querying specific infrastructural databases and protocols.

- **WHOIS:** A query and response protocol used for querying databases that store registered users or assignees of an Internet resource (domain name, IP block).
- **DNS Footprinting:** Extracting records (A, MX, NS, TXT) to understand how the target routes its traffic.
- **Email Footprinting:** Analyzing email headers to track the path an email took from sender to receiver, revealing internal IP schemes.
- **Network Footprinting:** Mapping out the logical network topology using tools like traceroute to find firewalls and routers.

3. Detailed Procedure

Step 1: WHOIS Lookup

Use a command-line utility or web interface (like WHOIS.net) to find domain registration details.

```
Command: whois example.com
```

Step 2: DNS Footprinting

Use `nslookup` or `dig` commands to query domain name servers and extract specific records.

```
Command: dig example.com ANY
Command: nslookup -type=MX example.com
```

Step 3: Network Footprinting (Traceroute)

Map the path packets take to reach the target domain using ICMP or UDP packets.

```
Command (Linux): traceroute example.com
Command (Windows): tracert example.com
```

Step 4: Email Header Analysis

Obtain an email originating from the target domain, view the raw headers, and analyze the "Received:" lines to trace internal routing.

4. Simulated Observation & Outputs

4.1 WHOIS Output

```
Domain Name: EXAMPLE.COM
Registry Domain ID: 2336799_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.iana.org
Creation Date: 1995-08-14T04:00:00Z
Registry Expiry Date: 2025-08-13T04:00:00Z
Registrant Organization: Internet Assigned Numbers Authority
```

4.2 DNS Records

Record Type	Target / Data	Inference
A Record	198.51.100.25	IPv4 Address of the main web server.
MX Record	mail.example.com (Priority 10)	Mail Exchange server for incoming emails.

Record Type	Target / Data	Inference
NS Record	ns1.example.com, ns2.example.com	Authoritative Name Servers managing the domain.
TXT Record	"v=spf1 include:_spf.example.com ~all"	Sender Policy Framework configuration for email validation.

4.3 Network Trace (Traceroute)

```
traceroute to example.com (198.51.100.25), 30 hops max
 1  gateway.local (192.168.1.1)  1.2 ms
 2  isp-router-ext.net (203.0.113.1)  12.5 ms
 3  core-backbone-01.net (198.51.100.1)  25.4 ms
 4  firewall.example.com (198.51.100.10)  45.1 ms
 5  web.example.com (198.51.100.25)  46.8 ms
```

4.4 Email Header Insights

Analysis of a sample email header showed:

- **Originating IP:** 10.0.5.15 (Reveals the target uses a 10.0.0.0/8 private addressing scheme internally).
- **Mail Client:** Microsoft Outlook (Indicates Windows enterprise environment).

5. Result & Conclusion

Detailed technical footprinting successfully extracted the foundational architecture of the target network. By querying WHOIS, DNS, and mapping the network routes, a comprehensive overview of the target's external-facing assets and routing paths was documented.